

# Computational Methods for Single-Cell RNA-Seq Analysis of *Arabidopsis thaliana*

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*Chapter 2 - WT vs pifq tutorial: pseudobulk differential expression and functional enrichment*

SingleCell Pipeline

2026-07-15

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## Purpose

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Chapter 2 starts from the annotated object produced in Chapter 1 and runs pseudobulk differential expression by cell type. For this WT/pifq draft, the analysis intentionally uses the direct bibliography annotation column `celltype`. It does not use `celltype_grouped` and does not use `celltype_curated`.

The WT/pifq test currently has one Cell Ranger library per condition. Therefore the pseudo-replicates used here are a practical demonstration for method development and figure generation, not independent biological replicates. This limitation should remain visible in the final methods text.

## Step 1 - Start from Chapter 1 celltype annotation

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The input is the Chapter 1 annotated Seurat object. The object is loaded before grouping or manual curation, so the active annotation column is `celltype`. This keeps numeric labels such as 2, 4, 5, and 7 when no bibliography marker match was assigned.

```
R
PIPELINE_DIR <- Sys.getenv("PIPELINE_DIR", unset = "/workspace/workflow")
DATA_DIR     <- Sys.getenv("DATA_DIR",      unset = "/workspace")
base_dir     <- file.path(DATA_DIR, "resultados_wt")

source(file.path(PIPELINE_DIR, "load_libraries.R"))
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))

set.seed(1807)
setwd(DATA_DIR)
list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)

pbmc_harmony <- readRDS(file.path(dir_objects,
                                   "pbmc_harmony_annotated.rds"))

pseudobulk_annot_col <- "celltype"
pseudobulk_conditions <- c("WT", "pifq")
comparison_tag <- "WT_vs_pifq"
```

## Step 2 - Celltype and condition check

The first diagnostic counts cells per celltype and condition. Cell types with fewer than 20 cells in either WT or pifq are not used for DESeq2 in this draft. With the current data, 10 cell-type labels pass this filter.

```
R
min_cells_per_condition <- 20

celltype_condition_counts <- as.data.frame.matrix(
  table(pbmh_harmony[[pseudobulk_annot_col]][, 1],
        pbmh_harmony$condition)
)
celltype_condition_counts$celltype <- rownames(celltype_condition_counts)
celltype_condition_counts <- celltype_condition_counts[
  , c("celltype", pseudobulk_conditions)
]

eligible_celltypes <- celltype_condition_counts$celltype[
  apply(celltype_condition_counts[, pseudobulk_conditions, drop = FALSE],
        1, function(x) all(x >= min_cells_per_condition))
]
```



Cells per direct celltype annotation and condition.

## Step 3 - Build celltype subsets

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One Seurat subset is created per direct `celltype` label. Labels are only sanitized for object names and filenames; they are not grouped or manually curated.

```
R
cell_type_subsets <- create_cell_type_subsets(
  pbmc_harmony,
  annot_col = pseudobulk_annot_col
)

eligible_subset_names <- gsub("[^[:alnum:]]_", "_", eligible_celltypes)
cell_type_subsets <- cell_type_subsets[
  names(cell_type_subsets) %in% eligible_subset_names
]
```

The retained labels were: Epidermis Hypocotyl.1, Epidermis Cotyledon, Mesophyll, Epidermis Hypocotyl.2, PHLOEM, ENDO1, 2, 4, 5, and 7.

## Step 4 - Pseudo-replicate assignment

Cells inside each retained cell type are randomly assigned to three pseudo-replicates per condition. This creates replicate-level count columns for DESeq2. Because the dataset has one library per condition, these pseudo-replicates are used to exercise the workflow and should not be described as biological replicates.

*R*

```
n_pseudoreps <- 3

cell_type_subsets_replicates <- assign_pseudoreplicates_batch(
  cell_type_subsets,
  pseudobulk_conditions = pseudobulk_conditions,
  n_pseudoreps = n_pseudoreps
)
```



Cells assigned to each pseudo-replicate for retained cell types.

## Step 5 - Pseudobulk tables and DESeq2

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Counts are aggregated by pseudo-replicate for each direct celltype label. DESeq2 is then run with WT as the reference and pifq as the treatment, so positive log2FC values indicate higher expression in pifq.

```
R
comparisons <- list(
  list(conds = c("WT", "pifq"), tag = "WT_vs_pifq")
)

deseq2_results <- run_pseudobulk_deseq2_analysis(
  cell_type_subsets_replicates = cell_type_subsets_replicates,
  comparisons = comparisons,
  output_dir = dir_06,
  cell_types = NULL,
  pseudobulk_dir = file.path(dir_objects,
                             "pseudobulk_replicas_celltype")
)
```

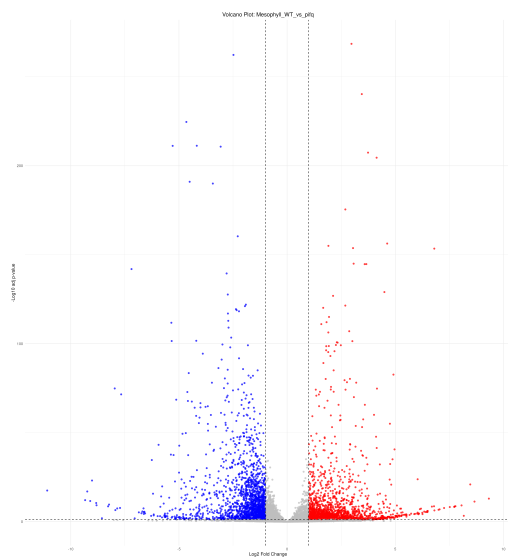
This run produced one DESeq2 result file for each of the 10 retained direct celltype labels under resultados\_wt/06\_de\_results/WT\_vs\_pifq/.

## Step 6 - Volcano plots

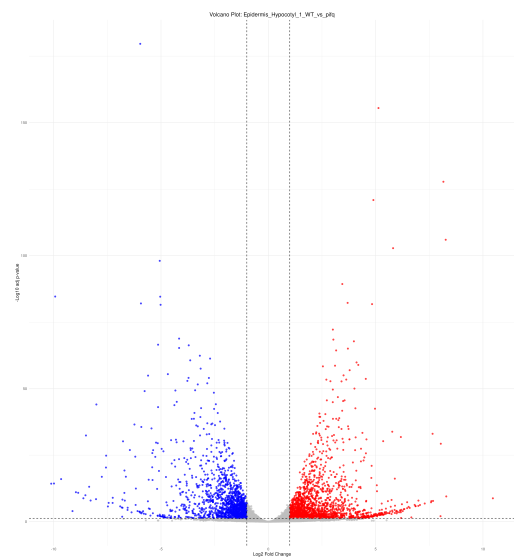
Volcano plots are rendered per cell type with adjusted p-value threshold 0.05 and absolute log2FC threshold 1. The examples below show Mesophyll and Epidermis Hypocotyl.1.

```
R
volcano_tag <- "WT_vs_pifq"
padj_cut <- 0.05
lfc_cut <- 1

render_volcano_plots(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag, "volcano"),
  pdf_name    = paste0("VolcanoPlots_", volcano_tag, ".pdf"),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut
)
```



Representative volcano plot: Mesophyll.



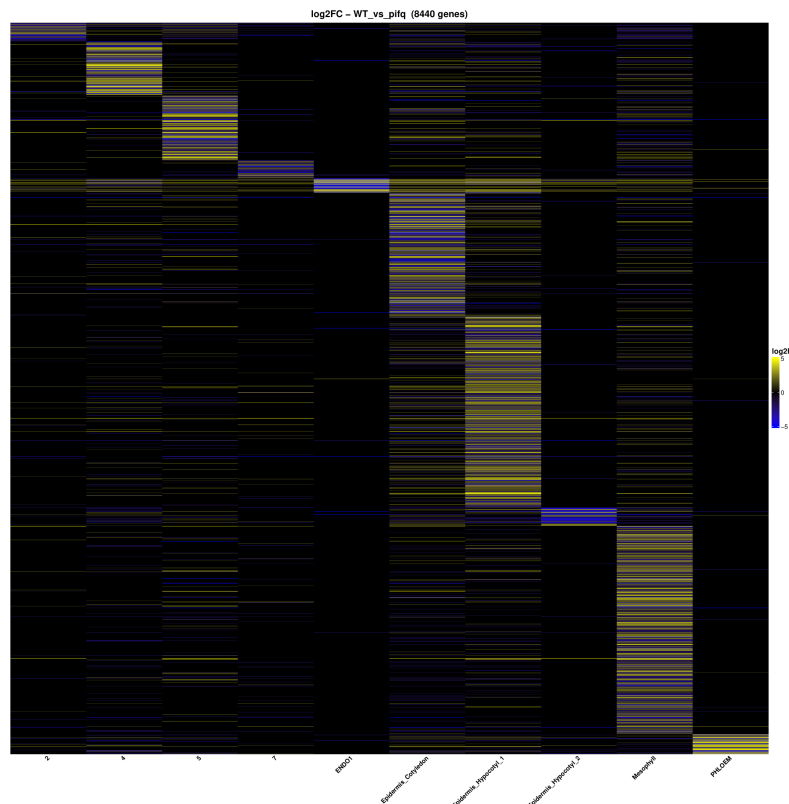
Representative volcano plot: Epidermis Hypocotyl.1.

## Step 7 - Differential tables and log2FC heatmap

Per-cell-type DESeq2 outputs are combined into a differential-gene matrix and a log2FC matrix. The heatmap uses the same staircase logic as the helper function: genes are ordered by the cell type where the absolute log2FC is largest. The current WT/pifq draft includes 8,440 genes after filtering with adjusted p-value 0.05 and absolute log2FC 1.

```
R
diff_tables <- build_differential_tables(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut,
  prefix      = paste0("diff_table_", volcano_tag)
)

build_logfc_heatmap(
  logfc_table = diff_tables$logfc,
  contrast_tag = volcano_tag,
  output_dir  = file.path(dir_06, volcano_tag),
  limits      = c(-5, 5)
)
```



Combined log2FC heatmap across retained direct celltype labels.

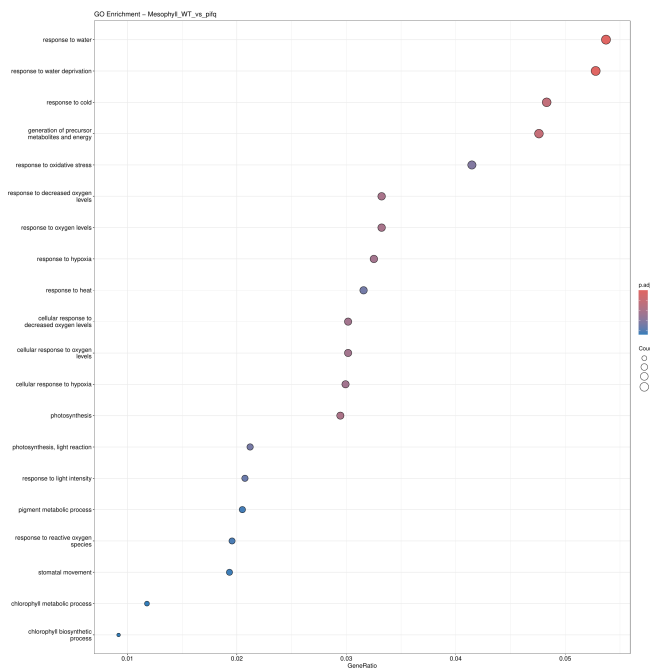


## Step 8 - GO enrichment

GO enrichment is run per retained cell type from significant genes in the WT vs pifq DESeq2 outputs. The Arabidopsis OrgDb annotation is used with TAIR gene identifiers and biological process ontology.

*R*

```
go_results <- run_go_enrichment_for_contrast(  
  results_dir = file.path(dir_06, volcano_tag),  
  output_dir  = file.path(dir_07, volcano_tag),  
  orgdb       = org.At.tair.db,  
  keytype     = "TAIR",  
  go_space    = "BP",  
  padj_cutoff = 0.05,  
  contrast_tag = volcano_tag  
)
```



Representative GO biological-process bubble plot for Mesophyll.

## Outputs and next decisions

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The tested Chapter 2 script is `workflow/capitulo2_wt_pifq.R`. Main outputs are written under `resultados_wt/06_de_results/WT_vs_pifq/` and `resultados_wt/07_go/WT_vs_pifq/`. The next decision is whether the final methods paper should stop at DESeq2/GO for this dataset or continue into the optional `hdWGCNA/network` sections after the pseudo-replicate limitation is addressed in the text.